

Postpartum Hemorrhage 02: The First Five Minutes

Companion reading handout for simultaneous listening.

Companion reading handout for simultaneous listening.

The first five minutes: team response.

In episode one, we built the mental model.

PPH is dangerous because the woman can lose blood faster than her vital signs admit, and because the uterus is supposed to be the tourniquet.

Now we move from recognition to action

This episode is about the first five minutes.

Not the first five minutes as a list.

The first five minutes as choreography.

Because in a real hemorrhage, the problem is usually not that nobody knows anything. The problem is that everyone is doing something, but not enough people are doing the right things in parallel.

One person is massaging the uterus, but nobody has called anesthesia.

Someone is looking for oxytocin, but nobody has sent blood.

Someone is documenting blood loss, but nobody is inspecting the cervix.

Someone says, "She is stable," but nobody has noticed that the blood loss is now crossing the threshold where the operating room, Bakri balloon, embolization, and surgery must enter the conversation.

So here is the anchor for this lecture.

One person at the uterus.

One person at the veins.

One person at the phone.

One person thinking.

That is the first five minutes.

One person at the uterus.

One person at the veins.

One person at the phone.

One person thinking.

Let us start at the bedside

A woman has just delivered. The placenta is out. Bleeding is heavier than expected. The pad is soaking. The uterus is soft. The midwife says, "I think she is bleeding.

You walk in.

Your first job is not to look calm.

Your first job is to create order.

The room needs to know that this is now a PPH response.

You say it out loud.

This is postpartum hemorrhage. I want help in the room now.

That sentence matters.

It changes the physiology of the team.

It gives permission to stop routine care and begin emergency care.

It tells the midwife that the quantification of blood loss matters.

It tells nursing that access and bloods are not optional.

It tells the senior that this is not a private problem for the junior physician.

It tells anesthesia that a maternal resuscitation may be developing.

And it tells you to stop thinking in a single line.

The first principle is parallel action.

Do not finish uterine massage before calling help.

Do not wait for the complete blood count before ordering blood products.

Do not wait for hypotension before placing large IV lines.

Do not give a second uterotonic and then start wondering whether the cervix was inspected.

The protocol and the position paper both make the same practical point: the order shown on paper is not a sacred sequence. The treatment bundle has to happen at the same time.

So, in the first minute, your voice should assign jobs.

You say to the midwife:

Stay at the uterus. Massage the fundus. Tell me if the uterus firms or stays soft.
Quantify blood loss.

You say to the nurse:

Two large-bore IV lines. Send type and crossmatch, complete blood count, coagulation profile, and fibrinogen. Start fluids.

You say to another staff member:

Call the senior obstetrician, call anesthesia, notify the operating room, and notify the blood bank that we have a possible postpartum hemorrhage.

You say to yourself, or out loud if you can:

Tone, trauma, tissue, thrombin.

And then you do not let the room drift.

This is where leadership is very practical.

Leadership is not a speech.

Leadership is making sure uterus, veins, phone, and thinking are all owned.

Let us take each track.

Track one is the uterus.

If the uterus is soft, the most common cause is atony. Atony is common enough that your hands should go there immediately.

Fundal massage.

Bimanual compression if needed.

Ask whether the bladder is full.

Empty the bladder.

Insert a Foley catheter.

This is not a minor detail.

A distended bladder can prevent the uterus from contracting well. A Foley catheter also gives you urine output, and urine output is a resuscitation monitor. In major hemorrhage, oliguria is not a laboratory abnormality. It is the kidney telling you that perfusion is failing.

So the uterus track sounds like this:

Massage the uterus. Empty the bladder. Insert Foley. Tell me the urine output.

If the uterus becomes firm and bleeding slows, you have likely found the major source.

But do not stop thinking.

A firm uterus with continuing bleeding means keep looking.

Trauma may be bleeding below your hand.

Retained tissue may be preventing full contraction.

Coagulopathy may be converting a controlled bleed into uncontrolled oozing.

Track two is the veins.

The local protocol asks for two 18 gauge IV lines in stage one, with a 14 gauge line if necessary after local anesthesia.

The checklist uses large access as an early action.

The exact gauge is less important than the principle: small, unreliable access is not acceptable in real hemorrhage.

You need large-bore access before she looks shocked.

The veins track sounds like this:

Two large-bore IV lines now. Draw blood before fluids if you can, but do not delay resuscitation. Send type and crossmatch, complete blood count, coagulation tests, and fibrinogen.

Fibrinogen deserves special mention.

In obstetric bleeding, fibrinogen can fall early and can predict severity. If you wait until the bleeding has become obviously catastrophic, you are already behind the coagulation problem.

So include fibrinogen in the first blood draw.

Also consider renal and liver function tests when the clinical context requires them, especially when there is preeclampsia, severe disease, or evolving shock.

Start fluids.

Use warmed crystalloid when possible.

But say this carefully in your mind: fluid is a bridge, not the destination.

Crystalloid can buy time while blood products are prepared, but a hemorrhaging patient ultimately needs oxygen-carrying capacity, coagulation factors, platelets when indicated, temperature control, and source control.

If the bleeding is significant, the blood bank needs to know early.

That brings us to track three: the phone.

The phone is not clerical.

The phone is treatment.

Calling help is not an admission that you have failed. Calling help is source control preparation.

The position paper explicitly emphasizes team activation: midwives, nurses, anesthesiologists, blood bank, and laboratory personnel.

The local protocol allows full or partial activation according to the physician in charge. Full activation can include the laboratory, anesthesia, operating room, and hematology.

If the hemorrhage is in a ward rather than in the delivery room, the ward resident must be summoned quickly and the senior on call updated.

So the phone track is concrete.

Call the senior obstetrician.

Call anesthesia.

Notify the operating room.

Notify the blood bank.

Tell the lab this is a postpartum hemorrhage.

Do not say, "Can someone call someone?"

Name the person.

Dana, call anesthesia and tell them: postpartum hemorrhage, ongoing bleeding, possible need for operating room.

Maya, call blood bank and tell them: postpartum hemorrhage, crossmatch sent, prepare two packed cell units and two fresh frozen plasma units.

You want closed-loop communication.

The person repeats the task.

They come back with the answer.

If nobody comes back, you ask again.

This is not about being dramatic. It is about preventing silent gaps.

In a hemorrhage, the dangerous delays are often invisible.

The blood sample was never sent.

The blood bank was never called.

Anesthesia heard about it only when the patient arrived unstable.

The operating room was not prepared because everyone assumed someone else had called.

The first five minutes should close those gaps.

Track four is thinking.

This is the easiest track to lose.

When the room becomes loud, the thinking physician can become just another pair of hands.

Sometimes that is necessary for a moment.

But someone must keep the whole frame in view.

What is the estimated blood loss?

Are the vital signs normal, or only still normal?

Is the bleeding ongoing?

Is the uterus soft or firm?

Was the placenta complete?

Was there an operative vaginal delivery, a difficult cesarean, a uterine scar, an extension, a suspected accreta, preeclampsia, abruption, fever, or prolonged labor?

Is the bleeding brisk red arterial bleeding, diffuse oozing, concealed abdominal bleeding, or uterine atony?

What stage are we in?

The four T's are not a memory trick at this point. They are your diagnostic dashboard.

Tone.

Is the uterus atonic? Is it enlarged? Is it responding to massage and uterotonics?

Trauma.

Inspect the cervix, vagina, vulva, perineum, and surgical field when relevant. Think about hematoma. Think about uterine rupture. In cesarean birth, think about extension of the incision or injured vessels.

Tissue.

Is placenta or membrane retained? Is the placenta complete? Could there be accreta?

Thrombin.

Is there a coagulation disorder? Is there abruption, severe preeclampsia, sepsis, amniotic fluid embolism, massive bleeding, or a falling fibrinogen?

This is the reason the first five minutes cannot be only uterotonics.

Uterotonics treat tone.

They do not repair a cervical tear.

They do not remove retained placenta.

They do not reverse DIC.

They do not fix uterine rupture.

So while one person treats tone, someone has to keep looking for the cause.

Now let us attach the local stage framework

Stage one is bleeding above the first threshold with normal vital signs

After vaginal birth, that means more than 500 milliliters. After cesarean birth, that means more than 1000 milliliters.

This is the window where a team can still be ahead.

Do not hear "normal vital signs" as comfort.

Hear it as time.

Normal vital signs in Stage one mean you have a chance to organize before compensation fails.

Stage one actions include calling for help, ensuring the uterus is contracted with bimanual or fundal massage, inserting two 18 gauge IV lines, giving fluids, placing a urinary catheter, drawing type and crossmatch, complete blood count, coagulation profile, and fibrinogen, ordering two packed cell units and two fresh frozen plasma units, giving uterotonic treatment, adding tranexamic acid, and examining the birth canal and uterine cavity as indicated.

That is a lot.

But in the room, it is simpler if you keep the four tracks.

Uterus.

Veins.

Phone.

Thinking.

One person at the uterus.

One person at the veins.

One person at the phone.

One person thinking.

Stage two is not just "a bit more bleeding"

Stage two is the point where you should be actively moving toward advanced control

The trigger is bleeding over 1000 milliliters after vaginal birth, over 1500 milliliters after cesarean birth, or continued bleeding after two uterotonics while vital signs are still normal.

That last phrase is important.

Continued bleeding after two uterotonics with normal vital signs is still Stage two.

Do not wait for hypotension to admit that the first-line plan is not enough.

In Stage two, the conversation changes.

The senior must be updated.

Additional access and repeat labs may be needed.

The patient should be transferred to the operating room for further treatment.

If atony is the cause and tears have been excluded, a Bakri balloon should enter the conversation.

If she is stable with persistent but slow bleeding, embolization may be considered according to medical judgment and local availability.

Transfusion should begin according to the protocol, including packed cells and fresh frozen plasma.

Surgical treatment must be on the horizon: laparotomy, compression sutures, uterine artery ligation, internal iliac ligation, and hysterectomy if needed to save life.

Notice the logic.

Stage two is still a stage where vital signs may be normal

But the treatment is already preparing for failure of simple measures.

That is excellent practice.

You do not wait until the patient is unstable to discover that you need the operating room.

Stage three is physiology failure until proven otherwise

The trigger is continued bleeding over 1500 milliliters, need for more than two blood units, suspected concealed bleeding or coagulopathy, abnormal vital signs, abnormal laboratory results, or oliguria.

This is the massive hemorrhage zone.

Activate the massive hemorrhage protocol.

Think source control and resuscitation together.

Replace blood and coagulation factors.

If DIC or hypofibrinogenemia is present, cryoprecipitate may be required.

Maintain oxygen saturation above 95 percent.

Prevent and correct hypothermia, acidosis, and coagulopathy.

During major transfusion, check calcium and potassium frequently, with the local protocol specifying every 15 minutes, and treat hypocalcemia and hyperkalemia aggressively.

If surgical control is not successful, hysterectomy must be considered.

The hard teaching point is this: Stage three is not the moment for optimism to slow the team.

You can be calm.

You can be respectful.

You can be surgically thoughtful.

But the physiology must be believed.

Ongoing hemorrhage with abnormal vital signs, oliguria, coagulopathy, or large transfusion requirement is not a situation where waiting is neutral. Waiting is an intervention, and it is usually the wrong one.

Stage four is shock, cardiac failure, or amniotic fluid embolism

At that point, advanced cardiac life support, the massive hemorrhage protocol, and surgical intervention continue as required.

For this course, the key is to understand how to avoid drifting into Stage four because the early tracks were not owned.

Now let us run the first five minutes as a script

You enter the room.

You see active bleeding.

You say:

This is postpartum hemorrhage. I need help now.

To the midwife:

Stay at the uterus. Fundal massage. Quantify blood loss. Tell me if the uterus is soft or firm.

To the nurse:

Two large-bore IV lines. Draw type and crossmatch, complete blood count, coagulation profile, and fibrinogen. Start warmed fluids.

To another staff member:

Call the senior obstetrician, anesthesia, operating room, lab, and blood bank. Tell blood bank we may need two packed cell units and two fresh frozen plasma units.

To the team:

Insert Foley. I want urine output.

Then you examine.

Uterus is soft. Start oxytocin. Add tranexamic acid. Keep massage.

Or:

Uterus is firm and bleeding continues. I am inspecting for trauma. Prepare for exam of cervix and vagina.

Or:

Placenta may be incomplete. I am concerned for tissue.

Or:

Bleeding is diffuse oozing. Send coags and fibrinogen urgently. Think thrombin.

Then you stage.

Estimated blood loss is 700 after vaginal birth, vital signs normal. This is Stage one. Continue the Stage one bundle.

Or:

Estimated blood loss is over 1000 after vaginal birth, still bleeding after uterotonics. This is Stage two. We are moving to the operating room. Prepare Bakri if atony after trauma is excluded.

Or:

Estimated blood loss is over 1500 with tachycardia and oliguria. This is Stage three. Activate massive hemorrhage protocol.

The words matter because they put the room in the same reality.

If you only say, "She is bleeding," people may still be operating at different levels of urgency.

If you say, "This is Stage two, we are moving to the operating room," the plan becomes visible.

If you say, "This is Stage three, activate massive hemorrhage," the physiology is named.

Now let us talk about the postpartum ward, because some of the most dangerous hemorrhages do not begin in a dramatic delivery room scene.

They begin quietly.

A nurse calls because the patient feels weak.

Or the uterus is not contracted.

Or the bleeding is increased.

Or the pulse is 112.

Or the blood pressure is below 90 over 50.

Or there is dyspnea, low saturation, dizziness, faintness, severe pain, abdominal distension after cesarean, bloody urine, or urine output below 35 milliliters per hour after cesarean.

The ward protocol is clear: report these findings, repeat and verify measurements when needed, perform clinical assessment, and obtain a complete blood count.

When calling a doctor for suspected increased postpartum hemorrhage, actions begin in parallel.

Start Hartmann's solution with 10 units of Pitocin.

Send a complete blood count.

Prepare equipment for an 18 gauge or larger IV.

Insert a Foley catheter.

Determine blood type and whether a suitable crossmatch sample exists in the blood bank.

If not, prepare the appropriate tube for sampling.

The key teaching point is that the ward is quieter, but the physiology is the same.

Do not let the quieter environment create a slower response.

The postpartum ward version of the four tracks is still:

Uterus.

Veins.

Phone.

Thinking.

Check uterine tone and bleeding.

Get access, blood count, and blood bank readiness.

Call the resident and update the senior.

Think through tone, trauma, tissue, and thrombin.

Now, before we close, let us sharpen the difference between activity and progress.

In hemorrhage, activity is easy.

People move.

Packages open.

Medications appear.

Vital signs are repeated.

Someone says, "We are doing everything.

But progress means four specific questions have answers.

First: Is the bleeding source being controlled?

Second: Is the circulation being supported with appropriate access, fluids, and blood products?

Third: Is the right help in motion?

Fourth: Has the team identified the cause, or at least kept all four causes actively in mind?

If one of those questions has no answer, the first five minutes are incomplete.

Let us finish with the core script you should be able to hear in your head.

This is postpartum hemorrhage.

Call help.

One person at the uterus.

One person at the veins.

One person at the phone.

One person thinking.

Massage the uterus, empty the bladder, insert Foley.

Two large-bore IV lines, type and crossmatch, complete blood count, coagulation profile, fibrinogen.

Notify senior obstetrician, anesthesia, operating room, lab, and blood bank.

Give oxytocin and tranexamic acid while we look for tone, trauma, tissue, and thrombin.

Stage the hemorrhage.

Stage one is the window to organize

Stage two means advanced control must enter the room

Stage three means massive hemorrhage and source control now

The first five minutes do not have to be elegant.

They have to be owned.

In the next episode, we will slow down and examine the medications: oxytocin, methylergometrine, carboprost, misoprostol, and tranexamic acid. We will focus on what each drug does, when to choose it, when to avoid it, and why medication is powerful but never enough when the cause is not tone.